Lecture 3 - Limits II

Garrett Credi

09/14/2026

Announcements

Homework Policy:

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- For homework assigned on Tueday, the due date will be Thursday @ 6PM

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Office Hours:

- Wednesday 12:15 - 1:15 PM
- Thursday 4:30 - 5:30 PM

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Both are in Vincent Hall 105.

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4. *Approximated* limits using a table of values
5. Computed limits by algebraic manipulation

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2. Discovering *limit properties* that help us reduce complicated limits to easier limits
3. Combining the above to get the process of algebraically computing limits

And we will also study cases where the limit is infinite.

Continuity

Continuous Functions I

To start off: let's consider the function:

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2. Using the graph of $f(x)$, what are:
   (1) $\lim_{x \to 3} f(x)$?
   (2) $\lim_{x \to 5} f(x)$?
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3. $\lim_{x \to a} f(x) = f(a)$

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We can define classes of functions that are specified to be continuous at important collections of points:

Continuous on $[a, b]$

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We can determine the points at which these functions are continuous (and thus the points at which it is easy to calculate limits) by discovering *limit properties*!

Limit Properties

Limit Properties: Additivity I

Say we have two functions $f(x)$ and $g(x)$, and we know that:

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1. As $x \to 0$, $f(x) \to 1$ and
2. As $x \to 0$, $g(x) \to 0$

When $x \to 0$, towards which value does the function $h(x) = f(x) + g(x)$ approach?

1. Property 1 says that as long as $x \approx 0$ then $f(x) \approx 1$

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So if $x \approx 0$ then $h(x) = f(x) + g(x) \approx 1 + 0 = 1$!

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Then we have that $\lim_{x \to a}(f(x) + g(x))$ exists and is equal to $b + c$.

Limit Properties: Multiplicativity

Multiplicativity of Limits

Suppose that $f(x)$ and $g(x)$ are two functions and a is a real number so that

Similarly:

Limits of Multiplicative Inverses

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Suppose that $f(x)$ is a function and a is a real number so that $\lim_{x \to a} f(x)$ exists, is equal to some real number b, and that $b \neq 0$.

Then we have that $\lim_{x \to a} \dfrac{1}{f(x)}$ exists

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Limit Properties: Composites

Limits of Composites

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Then we have that $\lim_{x \to a} g(f(x)) = c$.

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4. If $g(x)$ is instead continuous at $f(a)$ then $g(f(x))$ is continuous at $x = a$.

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The big upshot of this is the following:

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